Competitiveness, Cognitive Sophistication, and Speculative Behavior

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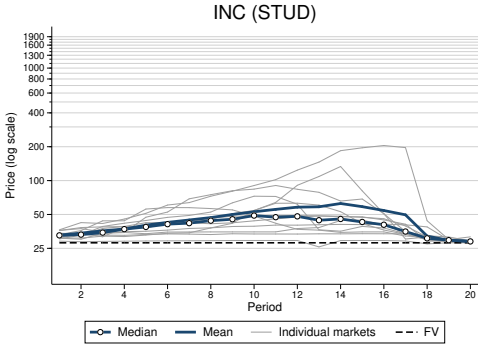
A bit of background

Weitzel et al. (2020, RFS): experimental asset markets with bubble formation



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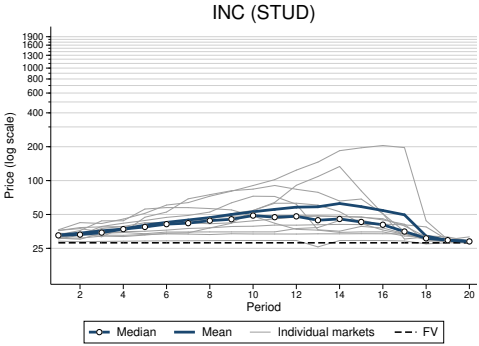

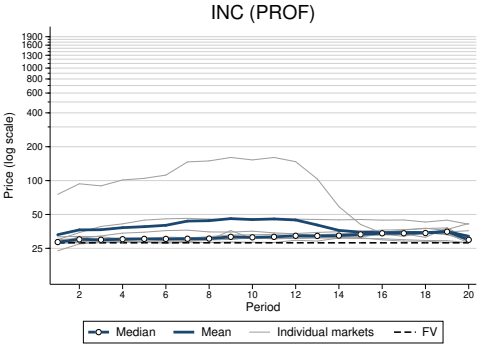

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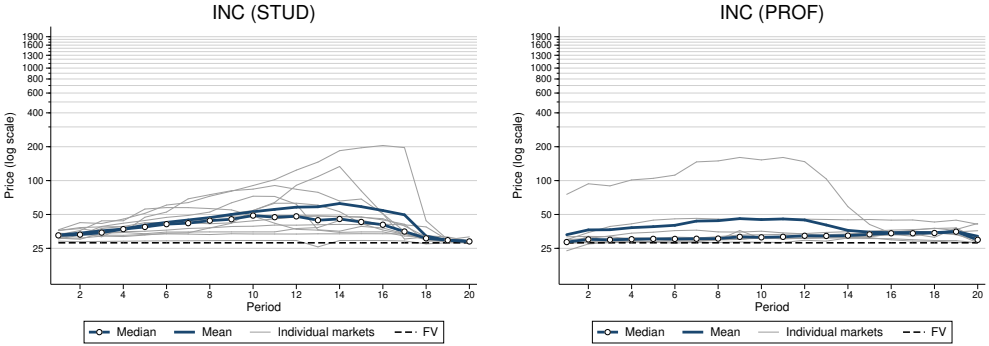

Why? What individual characteristics can explain speculative behavior?

Motivation

- Financial markets are inherently *competitive*
  - traders compete against each other
  - competitive business culture
  - tournament incentives

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→ Are these characteristics conducive for speculation?

→ Can common knowledge about all competitors having these same characteristics lead to more/less speculation?

Introduction

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    (e.g., Fellner/Maciejovsky, 2007; Breaban/Noussair, 2015; Eckel/Füllbrunn, 2015; Holt et al., 2017)
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- Competitiveness related to financial professionals
  (Kirchler et al., 2020)
  - → less speculation / fewer bubbles?
    (e.g., Weitzel et al., 2020; Cipriani et al., 2020

Introduction

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  - Cognitive sophistication related to market efficiency (fewer bubbles) and trading success (Bosch-Rosa et al., 2018; Corgnet et al. 2018)
  - Common knowledge about subjects' rationality has been shown to eliminate bubbles (Cheung et al., 2014)

Hypotheses

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Subjects' propensity to speculate is higher with common knowledge about being in a high-competitiveness group than with common knowledge about being in a low-competitiveness group.

Hypothesis 3

Subjects' propensity to speculate is lower with common knowledge about being in a high-cognitive-sophistication group than with common knowledge about being in a low-cognitive-sophistication group.

Experiment

Laboratory experiment with 3 steps:

1. Elicit subjects' individual willingness to compete / cognitive sophistication:
   - Task: add as many two-digits numbers as possible within 2 minutes

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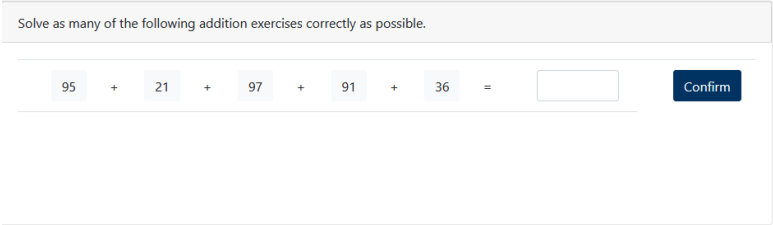

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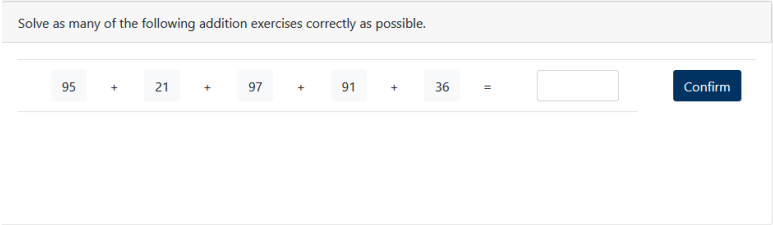

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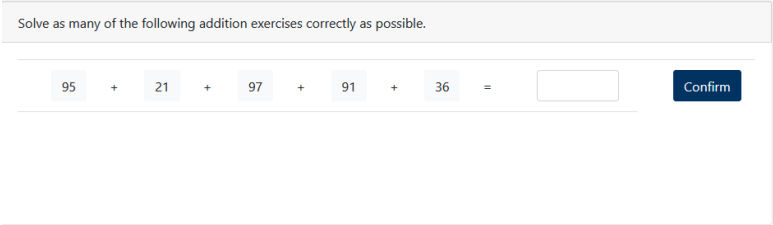

   - performance → **cognitive sophistication**
   - (i) piece-rate incentives; (ii) tournament incentives

     (iii) choice: percentage tournament incentives → **competitiveness**

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   - $COMP^{HIGH}$
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   - Information to subjects: "The three players in your group – including yourself – are all among the __ participants (50%) in this room who …"
     - "… invested the lowest/highest number of points into the Tournament payment in Stage 3."
     - " solved the lowest/highest number of calculations in Part 1."

Experiment

3. ’Bubble Game’ (Moinas/Pouget, 2013) within these groups of three:

   - sequential market for an asset with a fundamental value of 0
   - one by one, each trader is offered a price and given the possibility to “buy” or “not buy”
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- prices are exogenously given and are powers of 10:

Initial Price	Probability that this initial price is realised
1	50.00 %
10	25.00 %
100	12.50 %
1,000	6.25 %
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- E.g.:
  - $P_1 = 1$
  - $P_2 = 10$
  - $P_3 = 100$

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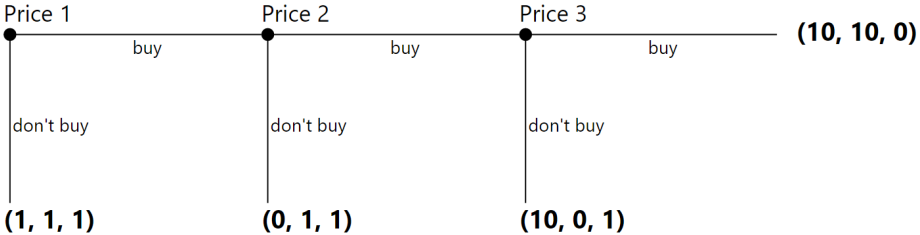

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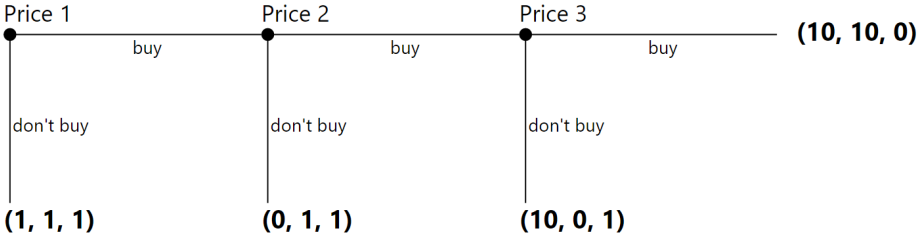

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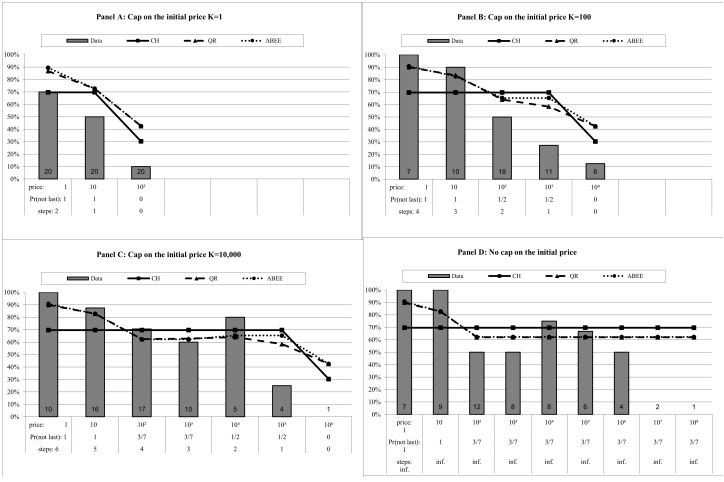

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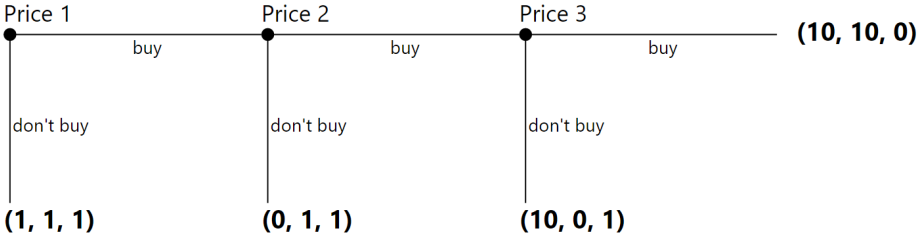

- Bayesian Nash Equilibrium: No-bubble, no one buys

- strategy method
- → individual measure for **propensity to speculate** (Janssen et al., 2019)

Experiment

- Five treatments:

	Information level		
	none	LOW-group	HIGH-group
Information about others' willingness to compete	BASE	PERF_{LOW}	$\text{PERF}^{\text{HIGH}}$
Information about others' performance		COMP_{LOW}	$\text{COMP}^{\text{HIGH}}$

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- 432 **male** participants so far (preregistered: 480, i.e., 96/treatment)
- Student participants, avg. age per lab: 22.4 - 24.7

- Since March 2022, data collection still ongoing in several labs: WU Vienna (222), Innsbruck (78), Graz (12), Brno (120)

Results

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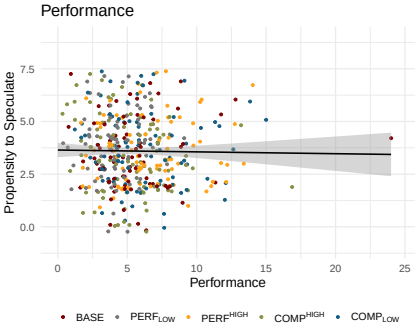
Performance
Propensity to Speculate
7.5
5.0
2.5
0.0
0
5
10
15
20
25
Performance
BASE
PERF$_{LOW}$
PERF^{HIGH}
COMP^{HIGH}
COMP$_{LOW}$

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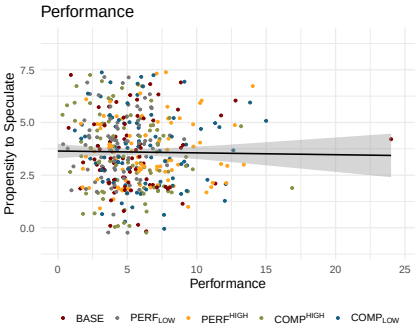

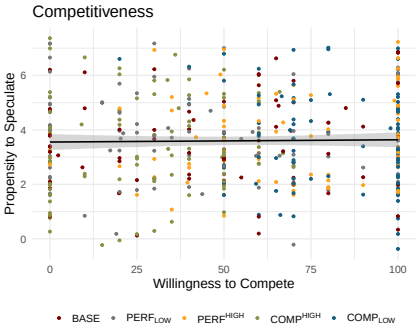

BASE: $\rho = 0.053$
PERF: $\rho = -0.099$
COMP: $\rho = 0.063$

$\rho = 0.036$
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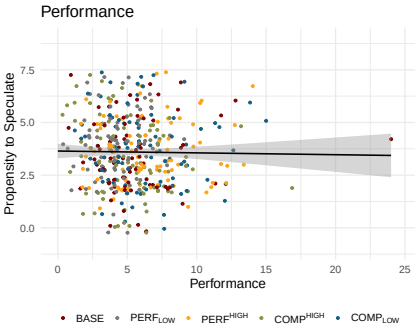

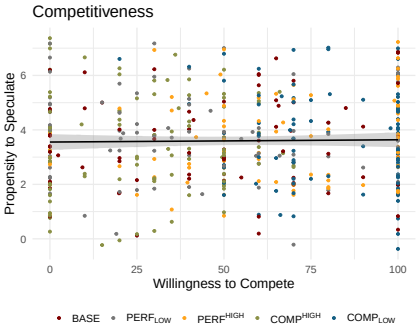

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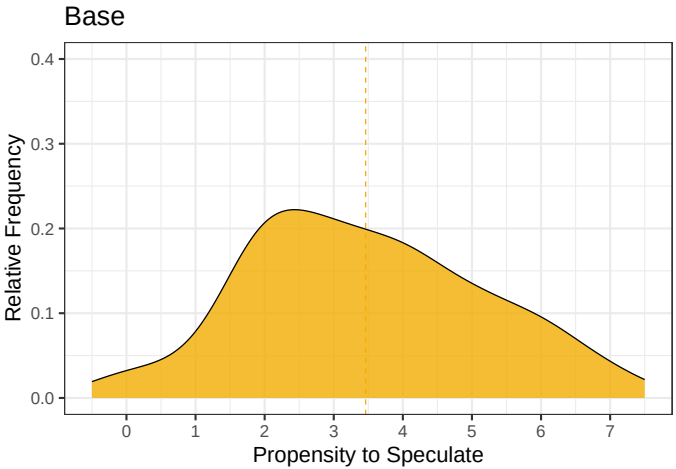
Corr. Performance $\longleftrightarrow$ Competitiveness:
$\rho = 0.241$ ($p < 0.01$)

Results

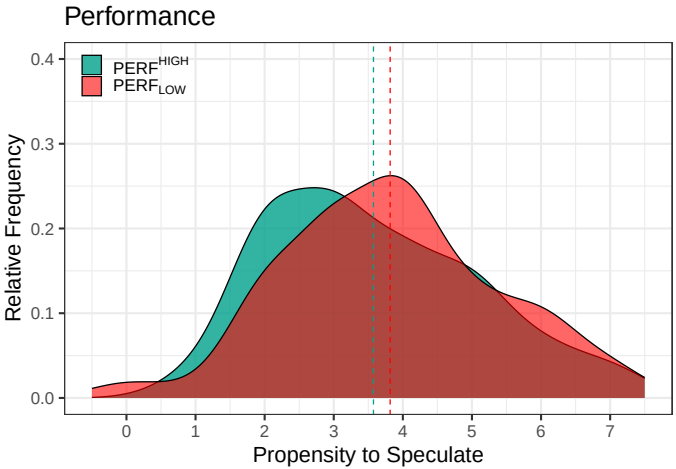
Propensity to speculate across treatments

Treatment	Mean	S.d.	Obs.
BASE	3.462	(1.680)	78
$\text{PERF}^{\text{HIGH}}$	3.575	(1.507)	87
PERF_{LOW}	3.816	(1.529)	87
$\text{COMP}^{\text{HIGH}}$	3.656	(1.717)	90
COMP_{LOW}	3.467	(1.794)	90

Results

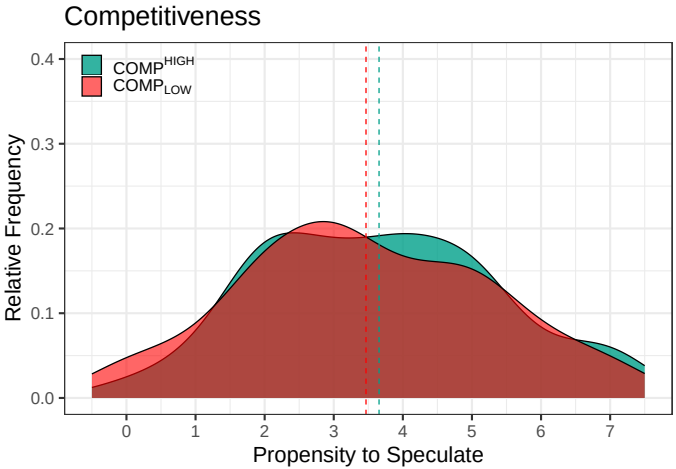

Base
Relative Frequency
0.0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
7
Propensity to Speculate

Results



t-test: $p = 0.296$
Wilcoxon ranksum test: $p = 0.200$

Results



t-test: $p = 0.471$
Wilcoxon ranksum test: $p = 0.543$

Results

What can explain speculation?

- Risk preferences (SOEP): $\rho = 0.252$ ($p < 0.01$)
- CRT (4 items): $\rho = -0.008$
- Overconfidence: $\rho = -0.057$

Results

	Dependent variable: Speculation		
	(1)	(2)	(3)
PERF_{LOW}	0.355	0.337	0.378
	(0.257)	(0.262)	(0.255)
$\text{PERF}^{\text{HIGH}}$	0.113	0.096	0.154
	(0.257)	(0.262)	(0.256)
COMP_{LOW}	0.005	−0.062	0.056
	(0.255)	(0.262)	(0.256)
$\text{COMP}^{\text{HIGH}}$	0.194	0.127	0.036
	(0.255)	(0.262)	(0.256)
CRT			0.017
			(0.068)
Age			−0.003
			(0.026)
Risk preferences			0.205***
			(0.038)
Constant	3.462***	3.426***	2.251***
	(0.187)	(0.228)	(0.714)
Lab FE	No	Yes	Yes
Observations	432	432	432
R^2	0.006	0.013	0.078
Adjusted R^2	−0.003	−0.003	0.057
Note:			*p<0.1; **p<0.05; ***p<0.01

Exploratory Results

Beliefs about *others'* cognitive soph. / competitiveness

- How smart or intelligent do you think the other participants in your group from Part 2 are? (Likert scale, 1-7)
- How willing to take part in a competition do you think the other participants in your group from Part 2 are? (Likert scale, 1-7)

Treatment	Spec.	Beliefs(Cog. Soph.)	Beliefs(Competitiveness)	N
BASE	3.462	4.603	4.487	78
$PERF^{HIGH}$	3.575	4.783***	$4.817^{\dagger\dagger}$	60
$PERF_{LOW}$	3.816	4.317***	$4.250^{\dagger\dagger}$	60
$COMP^{HIGH}$	3.656	4.725	5.101^{+++}	69
$COMP_{LOW}$	3.467	4.696	4.362^{+++}	69

Exploratory Results

Beliefs about *others'* cognitive soph. / competitiveness

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  → Spearman rank correlation: $\rho = 0.09$ ($p = 0.076$)

- How willing to take part in a competition do you think the other participants in your group from Part 2 are? (Likert scale, 1-7)

  → Spearman rank correlation: $\rho = 0.02$ ($p = 0.717$)

Conclusion

Competing to be the “greater fool”?

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- Common knowledge about others individual competitiveness
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- What (seems to) matter:
  - Beliefs about *others’* cognitive sophistication
  - Risk preferences

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